

Open



Practical10.c

/mnt/c/WINDOWS/system32/os/practical

```
1
2#include <stdio.h>
3#include <fcntl.h>
4#include <unistd.h>
5#include <sys/mman.h>
6#include <sys/stat.h>
7#include <string.h>
8
9int main() {
10    int fd = open("data.txt", O_RDWR | O_CREAT, 0644);
11
12    if (fd == -1) {
13        perror("open");
14        return 1;
15    }
16
17    const char *msg = "Hello using mmap";
18    size_t len = strlen(msg);
19
20    if (ftruncate(fd, len) == -1) {
21        perror("ftruncate");
22        close(fd);
23        return 1;
24    }
25
26    char *p = mmap(NULL, len, PROT_READ | PROT_WRITE,
27        MAP_SHARED, fd, 0);
28
29    if (p == MAP_FAILED) {
30        perror("mmap");
31        close(fd);
32        return 1;
33    }
34
35    memcpy(p, msg, len);
36    printf("File content: %.s\n", (int)len, p);
37
38    if (msync(p, len, MS_SYNC) == -1)
39        perror("msync");
40
41    munmap(p, len);
42    close(fd);
43
44    return 0;
45 }
```

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(gedit:9160): WARNING **: 03:18:31.283: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ echo "Operating Systems" > file.txt

ls -i file.txt
stat file.txt

ln file.txt hardlink.txt
ln -s file.txt symlink.txt

ls -li file.txt hardlink.txt symlink.txt
stat hardlink.txt
stat symlink.txt

find . -inum "$(ls -i file.txt | awk '{print $1}')"
86975767803640131 file.txt
  File: file.txt
  size: 18          Blocks: 0          IO Block: 4096   regular file
Device: 0,70   Inode: 86975767803640131   Links: 1
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/ sampath)   Gid: ( 1000/ sampath)
Access: 2026-10-07 03:17:54.342283000 +0000
Modify: 2026-10-07 03:17:54.342283000 +0000
Change: 2026-10-07 03:17:54.342283000 +0000
Birth: -
86975767803640131 -rwxrwxrwx 2 sampath sampath 18 Oct  7 03:17 file.txt
86975767803640131 -rwxrwxrwx 2 sampath sampath 18 Oct  7 03:17 hardlink.txt
34621422135458237 lrwxrwxrwx 1 sampath sampath  8 Oct  7 03:17 symlink.txt -> file.txt
  File: hardlink.txt
  size: 18          Blocks: 0          IO Block: 4096   regular file
Device: 0,70   Inode: 86975767803640131   Links: 2
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/ sampath)   Gid: ( 1000/ sampath)
Access: 2026-10-07 03:17:54.342283000 +0000
Modify: 2026-10-07 03:17:54.342283000 +0000
Change: 2026-10-07 03:17:54.433722800 +0000
Birth: -
  File: 'symlink.txt' -> 'file.txt'
  size: 8          Blocks: 0          IO Block: 4096   symbolic link
Device: 0,70   Inode: 34621422135458237   Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ sampath)   Gid: ( 1000/ sampath)
Access: 2026-10-07 03:17:54.451426900 +0000
Modify: 2026-10-07 03:17:54.451426900 +0000
Change: 2026-10-07 03:17:54.451426900 +0000
Birth: -
./file.txt
./hardlink.txt

```

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sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gedit Practical9.c

** (gedit:9160): WARNING **: 03:18:31.283: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:9160): WARNING **: 03:18:31.283: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gedit Practical10.c

** (gedit:9949): WARNING **: 03:18:45.741: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:9949): WARNING **: 03:18:45.741: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gcc Practical10.c -o Practical10
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ./Practical10
File content: Hello using mmap
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gedit Practical10.c

** (gedit:11474): WARNING **: 03:20:22.516: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:11474): WARNING **: 03:20:22.516: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$

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